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IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

MSc and EEE/EIE PART III/IV: MEng, BEng and ACGI

MATHEMATICS FOR SIGNALS AND SYSTEMS

Wednesday, 6 December 10:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions. All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : S. Parbhoo

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Special Information for the Invigilators: none

Information for Candidates:

For questions involving matrix manipulations or calculations, please provide answers showing intermediate calculations steps.

Given a matrix $\mathbf{A}$ with full column rank, the orthogonal projection onto the space spanned by the columns of $\mathbf{A}$ is given by: $\mathbf{P} = \mathbf{A}(\mathbf{A}^H\mathbf{A})^{-1}\mathbf{A}^H$.

The inverse of an upper triangular matrix is also an upper triangular matrix.

The Questions

1. (a) Consider the system $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{A} \in \mathbb{R}^{3\times 3}$ and $\mathbf{b} = [4, 8, 4]^T$. The complete solution to this system is

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + a \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + b \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

for arbitrary a and b.

i. Find the rank of $\mathbf{A}$ [2]

ii. Find $\mathbf{A}$ [2]

iii. Find the minimum norm solution. [2]

(b) Assume that $\mathbf{A} \in \mathbb{R}^{n\times n}$. If $\mathbf{Ax} = \mathbf{0}$ has a non-zero solution, show that $\mathbf{A}^T\mathbf{y} = \mathbf{b}$ fails to be solvable for some choices of $\mathbf{b}$. Construct an example of $\mathbf{A}$ and $\mathbf{b}$. [2]

(c) Determine the four fundamental subspaces of

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 4 & 6 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

[4]

Question 1 continues on the next page

(d) Consider the system $\mathbf{Ax} = \mathbf{b}$ with

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \end{bmatrix}$$

and $\mathbf{b} = [1, 7]^T$. A basis for the null space of $\mathbf{A}$ is $\mathbf{n} = [0, -1, 2]^T$ and a possible solution to the system is $\mathbf{x}_1 = [1, 2, 3]^T$. Find an alternative solution $\mathbf{x}_2$ such that $\|\mathbf{x}_2\| < \|\mathbf{x}_1\|$. [2]

(e) The linear combination of $\mathbf{v} = [1, 1, 0]^T$ and $\mathbf{w} = [0, 1, 1]^T$ fills a plane in $\mathbb{R}^3$.

i. Find a vector $\mathbf{z}$ that is perpendicular to the plane generated by $\mathbf{v}$ and $\mathbf{w}$. [2]

ii. Find a vector $\mathbf{u}$ that is not on the plane and that is different from $\alpha\mathbf{z}$ for an arbitrary constant α. [2]

(f) Show that if a matrix $\mathbf{S}$ is symmetric and orthogonal then $\mathbf{S}^2 = \mathbf{I}$. [2]

2. (a) Let

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & 1 \\ 4 & 12 & 4 \end{bmatrix}.$$

i. Find the matrix $\mathbf{P}_C$ that models the orthogonal projection onto the space spanned by the columns of $\mathbf{A}$. [3]

ii. Now, find the projection matrix $\mathbf{P}_R$ onto the row space of the above matrix. [3]

iii. Assume $\mathbf{x} = [22, 33, 11]^T$. Is the vector $\hat{\mathbf{x}} = [12, 36, 12]^T$ the orthogonal projection of $\mathbf{x}$ onto the space spanned by the rows of $\mathbf{A}$? Briefly justify your answer. [3]

(b) Consider again the matrix $\mathbf{A}$ of part (a).

i. Can you guarantee that the system $\mathbf{Ax} = \mathbf{b}$, has always a solution? Briefly justify your answer. [2]

ii. Can you guarantee that the system $\mathbf{Ax} = \mathbf{P}_C\mathbf{b}$, has always a solution? Briefly justify your answer. [2]

iii. Assume $\mathbf{b} = [1, 0]^T$, find a solution to $\mathbf{Ax} = \mathbf{P}_C\mathbf{b}$. [2]

(c) i. If $\mathcal{P}$ is a projection matrix, prove so is $\mathbf{I} - \mathcal{P}$, where $\mathbf{I}$ is the identity matrix. [5]

3. (a) Let V be the vector space consisting of all functions of the form

$$f(x) = \alpha + \beta x + \gamma x^2,$$

for any $\alpha, \beta, \gamma \in \mathbb{R}$. Consider the linear transformation

$$L = \frac{df}{dx} - f(x).$$

i. Find the matrix representing the linear mapping L with respect to the basis $\{1, x, x^2\}$. [3]

ii. Use the answer from part i. to find one solution to the following differential equation:

$$\frac{df}{dx} - f(x) = 3 - x + 2x^2.$$

[3]

(b) Find a basis for the following subspaces of $\mathbb{R}^4$

i. The vectors for which $x_1 = x_3 + 2x_4$ [1]

ii. The vectors for which $x_1 + x_2 = 0$ [1]

Question 3 continues on the next page

(c) Consider the matrix

$$\mathbf{Q} = \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ a & 0.7 & 0.5 \\ 0 & b & 0.3 \end{bmatrix}.$$

i. Find the values of a and b so that $\mathbf{Q}$ models a Markov or "transition" matrix. Briefly justify your answer. [3]

ii. Assume $\mathbf{Q}$ models the fraction of population that moves every year across three different towns. Find the steady state of $\mathbf{Q}$. [3]

iii. Assume the total population at the beginning of your observation is $80,000$. What is the population in each town at the steady state? [3]

(d) Calculate the singular values of the following matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 2 & 0 & 1 \end{bmatrix}.$$

[3]